Parametric Survival Analysis of Acute Myelogenous Leukaemia (AML) Remission Data

Group 5
Faida Niyonshimiye
Oliver Nalunkuma
Sylvestre Mukiza
Patrick Makina
Kutlo Kejang

Lecturer: Prof. Innocent Ngaruye
Course: Survival Analysis

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Outline

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Introduction

Acute Myelogenous Leukaemia (AML) is an aggressive blood cancer with highly variable patient outcomes following treatment [1]. Understanding the time-to-death distribution is critical for clinical planning and evaluating treatment effectiveness.

Dataset: The `myeloid` dataset from the `survival` R package contains data on 646 AML patients enrolled in a randomised clinical trial comparing a standard treatment arm against an experimental one.

Objective: Fit and compare multiple parametric survival distributions to the AML remission data, select the best-fitting model and use it to characterise the survival experience of the cohort through life functions and key survival probabilities.

Key variables:

- futime - follow-up time in days (survival time)
- death - event indicator (1 = death, 0 = censored)
- Covariates - sex, treatment arm, FLT3 mutation level, transplant time, complete remission time, relapse time

Data Exploration – Baseline Characteristics

Table 1: Baseline Characteristics and Survival Outcome Summary.

Characteristic	Level / Statistic	Summary	Missing, N (%)	Min – Max
Total Participants	N (%)	646 (100.0%)	0 (0.00%)	
Events (deaths)	N (%)	320 (49.54%)	0 (0%)	
Censored	N (%)	326 (50.46%)	0 (0%)	
Follow-up time (days)	Median (IQR)	817 (286.5–1733.25)	0 (0%)	4–2419
Sex	Female	361 (55.88%)	0 (0%)	
	Male	285 (44.12%)		
Treatment	Standard	317 (49.07%)	0 (0%)	
	Experimental	329 (50.93%)		
FLT3 mutation level	Low	149 (23.07%)	0 (0%)	
	Moderate	319 (49.38%)		
	High	178 (27.55%)		
Transplant time (days)	Median (IQR)	147 (105–244.25)	282 (43.65%)	24–1526
Complete remission (days)	Median (IQR)	39 (32–57)	192 (29.72%)	21–554
Relapse time (days)	Median (IQR)	283 (181.5–470)	420 (65.02%)	26–1905

Data Exploration – Variable Distributions

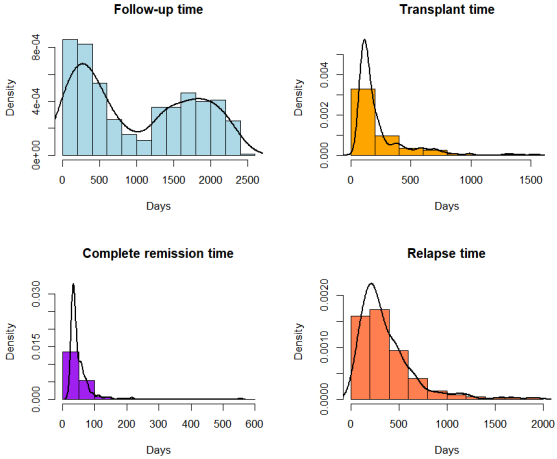

Figure 1: Density histograms of continuous variables.

Data Preparation

The dataset was prepared for parametric modelling in the following steps:

- **Data loading:** The `myeloid` dataset was loaded from the `survival` package via `data(cancer)`.
- **Factor recoding:** Treatment (`A/B` → Standard/Experimental), sex (`f/m` → Female/Male) and FLT3 (`A/B/C` → Low/Moderate/High) were relabelled for clarity.
- **Missing values:** No missing values were found in the `futime` and `death` variables.
- **Variable selection:** Only `futime` and `death` were retained into a cleaned dataset `myeloid_clean`.
- **Survival object:** Using the `survival` package, a `Surv(futime, death)` object was constructed as the response for all parametric model fits.

Model Fitting – Parametric Models

Seven parametric distributions were fitted to the survival object using maximum likelihood estimation (MLE) via the `flexsurv` R package.

Table 2: Parametric Model Comparison – MLE Parameter Estimates, Log-Likelihood and AIC.

Model	**Parameter 1**	**Parameter 2**	**Parameter 3**	**LogLik**	**AIC**
Exponential	rate = 0.0005	–	–	−2758.476	5518.952
Weibull	shape = 0.7029	scale = 2453.898	–	−2729.043	5462.087
LogNormal	meanlog = 7.238	sdlog = 1.9368	–	−2703.489	5410.978
LogLogistic	shape = 0.8799	scale = 1319.001	–	−2711.414	5426.829
Gamma	shape = 0.6689	rate = 0.0003	–	−2736.095	5476.191
GenGamma	mu = 6.6104	sigma = 2.1488	Q = −0.8558	−2696.550	5399.101
Gompertz	**shape** = −0.0015	**rate** = 0.0012	**–**	**−2682.729**	**5369.457**

Model Selection

The Akaike Information Criterion (AIC) was used to compare models, where lower AIC indicates a better balance between fit and parsimony. The **Gompertz model** achieves the lowest AIC of **5369.457**.

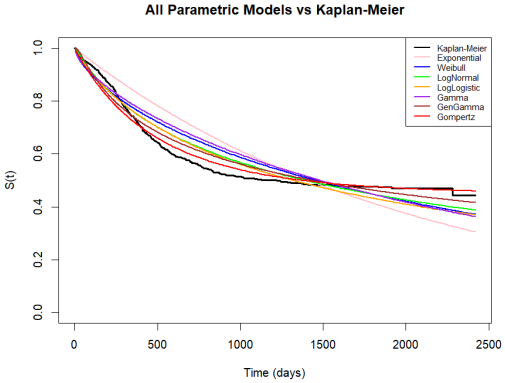

Figure 2: All parametric survival curves overlaid on the Kaplan–Meier estimate.

Life Functions – Gompertz Model

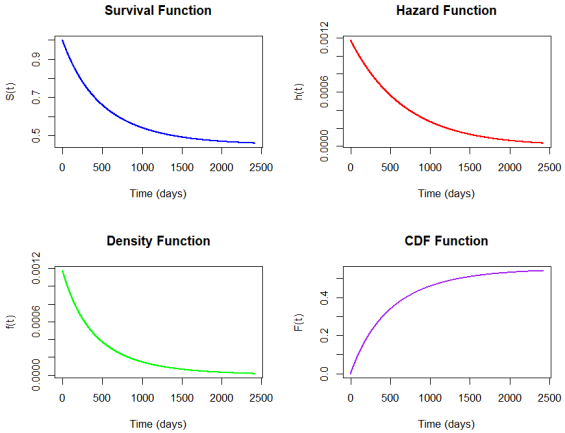

Figure 3: Life functions of the fitted Gompertz model.

Life Functions – Key Probabilities and Summary

Table 3: Survival Probabilities at Key Time Points.

Days	$S(t)$	$F(t)$	$h(t)$
30	0.9661	0.0339	0.001124
90	0.9057	0.0943	0.001029
180	0.8305	0.1695	0.000902
365	0.7176	0.2824	0.000687
730	0.5910	0.4090	0.000402
1095	0.5275	0.4725	0.000235

Table 4: Life Function Summary.

Metric	Value
Best Model	Gompertz
Mean survival (days)	271.6
Variance (days^2)	192,612.24

- **Survival probabilities** $S(365) = 0.7176$ indicates that 71.76% of patients survived beyond 1 year, while $S(1095) = 0.5275$ shows that just over half survived beyond 3 years.
- **Mean survival time of 271.6 days** ($\approx$ 9 months) lies well below the median follow-up of 817 days, because a concentration of early deaths pulls the mean down substantially.
- **Variance of 192,612 days^2** (SD $\approx$ 439 days) shows wide heterogeneity in outcomes, that is, patients respond very differently to treatment, leading to highly variable survival times.

Interpretation

Gompertz model: $h(t) = \alpha e^{\gamma t}, \quad \hat{\alpha} = 0.0012, \quad \hat{\gamma} = -0.0015$

- **Shape** ($\hat{\gamma} < 0$): A negative shape indicates an exponentially monotonic decreasing hazard, that is, the risk of death diminishes over time. This is clinically consistent with AML, where the highest risk of death occurs early due to treatment toxicity, infection and disease relapse, and patients who survive this critical window experience a steadily declining hazard.
- **Rate** ($\hat{\alpha} = 0.0012$ per day): At the start of follow-up, roughly 1 in every 646 patients is expected to die on any given day. By day 365 this drops to about 1 in every 1,000, and by day 1095 to roughly 1 in every 4,000, which confirms a steadily declining hazard over time.

Conclusions and Recommendations

Conclusions

- Among seven parametric models, the **Gompertz model** best fits the AML data (AIC = 5369.457), confirmed visually against the Kaplan-Meier curve.
- The **decreasing hazard** confirms that patients who survive the early post-treatment period face a progressively better prognosis.
- About **53%** of patients survived beyond 3 years, suggesting long-term remission is achievable for a meaningful proportion of the cohort.

Recommendations

- **Intensify early monitoring:** Since risk of death is highest in the first year, clinical resources should be concentrated in that window.
- **Incorporate covariates:** FLT3 mutation level, treatment arm and transplant timing should be included in a covariate-adjusted parametric model to quantify their individual effects on survival.
- **Extend the model:** Accelerated failure time (AFT) or proportional hazards (PH) parametric regression frameworks are natural next steps for this dataset.

References

Jennifer G Le-Rademacher, Ryan A Peterson, Terry M Therneau, Ben L Sanford, Richard M Stone, and Sumithra J Mandrekar.
Application of multi-state models in cancer clinical trials.
Clinical Trials, 15(5):489–498, 2018.

QnA

Thank You!

Questions and comments are welcome.